GlucoTwin. System architecture

AI PATIENT DIGITAL TWIN · 2-HOUR GLUCOSE FORECAST · HYPO / HYPER RISK WITH EXPLANATIONS

01 DATA SOURCES

EHR record
Demographics, diagnoses (SNOMED-CT), medications, HbA1c / eGFR labs (LOINC), prior hypos

CGM sensor
Glucose every 5 min; dropouts kept as gaps

Wearable + logs
Heart rate, steps, sleep; logged carbs and insulin

Real data (optional)
OhioT1DM XML, generic CSV, GlucoFM-Bench

`importers.py · glucofm.py`

`synthetic_data.py · one patient_id links all three`

02 STORAGE

SQLite store
`data_loader.py`

Patients, conditions, medications, labs and 5-min time series in one schema, keyed by patient_id.

PostgreSQL-ready: swap the connection for a SQLAlchemy engine.

Every row carries provenance = simulated; the hidden "true" glucose is stored apart and never used by any model.

Exports
FHIR R4-style JSON (SNOMED-CT / LOINC codes) and CSV, tagged synthetic.

03 PREPROCESSING

Preprocessing
`preprocessing.py`

Validate schema and value ranges

Resample to a regular 5-min grid

Interpolate CGM gaps ≤ 30 min

Longer gaps stay missing, never targets

Observed / imputed / missing flags on every sample

04 CAUSAL FEATURES

Glucose history
Lags, deltas, rolling mean / SD, time of day

Wearable & lifestyle
Heart rate, steps, sleep, carbs and insulin on board

EHR context
Diabetes type, duration, HbA1c, BMI, therapy, prior hypos

`features.py · only data ≤ t`

feature table

05 AI / ML MODELS

Glucose forecaster
`forecasting.py`

30 / 60 / 90 / 120 min; ridge vs XGBoost; split-conformal 80% interval

Hypo / hyper risk
`risk_prediction.py`

< 70 or > 250 mg/dL within 2 h; XGBoost + isotonic calibration; refuses when events are rare

Explainability
`explainability.py`

TreeSHAP / exact linear, grouped by data source

fitted models

06 DIGITAL TWIN

PatientTwin
`digital_twin.py`

EHR + sensor layers under one patient identity

Has a clock: every 5 min re-reads only data ≤ now and re-runs all models

Withholds forecasts if CGM is > 15 min stale

Tracks data freshness, imputation and missing inputs

What-if sensitivity for carbs / activity

07 CLINICIAN UI

Streamlit dashboard
`app.py`

Overview · Forecast

Clinical context

Risk & explainability

Model performance

Twin timeline replay

Patient journey · About

Clinician review
Informational only, no dosing advice

test metrics

Offline training & evaluation
`train.py · pipeline.py · evaluation.py`

SPLIT
30 training / 10 held-out patients, plus a chronological cut-off with a 2-h embargo

SELECTION
Model family and calibration chosen on a calibration split, never on test data

METRICS
MAE, RMSE, interval coverage, PR-AUC, Brier score, ECE

ABLATION
Glucose → + wearable → + EHR, 95% block-bootstrap CI

ARTIFACTS
joblib, fingerprinted; retrains automatically when data or config change

Glucose features · Wearable & lifestyle · EHR context · Live twin loop (every 5 min) · Data flow

RESEARCH PROTOTYPE · SYNTHETIC DATA · NOT A MEDICAL DEVICE